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A Review of Nigerian Bentonitic Clays as Drilling Mud

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Abstract

The heydays of bentonitic clay production in Nigeria occurred in the late 1950s when oil was first discovered. This was short lived as the advent of imported bentonite took the centre stage in the early 1960s. Since then, there has been no widespread use of the Nigerian bentonitic clay as drilling muds in Nigeria. In the recent past, precisely in 2001, the only major use of Nigerian bentonite for drilling oil wells was made by Shell Petroleum Development Company, Nigeria where over 600 wells were drilled with it. In recent years, Nigerian researchers have been actively pursuing the study of using Nigerian clay as drilling muds and finding potential markets for it. Many of these studies have focused on characterizing and beneficiating the clays to the American Petroleum Institute (API) standard. The available studies in this direction have been chronicled in this paper with the major research findings of each individual researcher brought to the fore. Much of the information that has been gathered in this review indicate that all the researchers were in unison in their submission that the Nigerian bentonitic clay is predominantly Calcium based and that it requires some measure of beneficiation to be effective for use as drilling muds. Mention is also made of possible gray areas where more research efforts should be directed. This work is predicated upon the belief that the works of the past combined with that of the present can lay the necessary framework for future studies on Nigerian bentonitic clays. The review this paper presents can serve as a baseline for the oil and gas E & P companies desirous of unlocking the potential of using Nigerian bentonitic clays as drilling muds.

Keywords : Bentonite, Nigerian clays, Beneficiation, Drilling mud

Introduction

As the world leans on oil and gas as the major source of energy, industry operators, in a bid to satisfy this demand have become positively restless. Today, as a result of their restlessness, oil and gas exploration and production activities dot virtually every part of the globe where hydrocarbon deposits are deemed to occur. In order to produce these hydrocarbons, the industry relies on a number of oilfield chemical formulations. One of such chemical formulations is the drilling fluid. In recent times, the demand for this fluid has gone sky high. Little wonder, in 2011, *Freedonia*, a market research organization reported that